THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE FULL SYLLABUS PAPER SET 09

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Citrus fruits such as lemon contain: 1

(a) lactic acid **(b)** citric acid **(c)** acetic acid **(d)** formic acid

2. A mixture of lemon juice and lime water in correct amounts becomes: 1

(a) red **(b)** more basic **(c)** neutral **(d)** more acidic

3. The bright part of a torch bulb is: 1

(a) terminal **(b)** base **(c)** casing **(d)** filament

4. Why are electrical wires covered with plastic? 1

(a) It makes them heavy
(b) Plastic is a conductor
(c) Plastic is an insulator and protects from shock
(d) It looks good

5. The hardest natural substance is: 1

(a) sodium **(b)** graphite **(c)** diamond **(d)** iron

6. The property of producing a ringing sound on striking is: 1

(a) brittleness **(b)** ductility **(c)** sonority **(d)** lustre

7. The gas formed when vinegar reacts with baking soda is: 1

(a) oxygen **(b)** hydrogen **(c)** nitrogen **(d)** carbon dioxide

8. Dissolving salt in water and recovering it shows: 1

(a) an irreversible chemical change **(b)** combustion
(c) a reversible physical change **(d)** rusting

9. During adolescence, a nutritious diet is important because: 1

(a) the body is growing rapidly **(b)** only sweets are needed
(c) we need less food **(d)** growth has stopped

10. If someone bullies you online, you should: 1

(a) ignore and say nothing ever **(b)** share it with strangers
(c) seek help from parents or teachers **(d)** bully back

11. The process by which a hot body gives heat to a cold body when they are in contact is: 1

(a) convection **(b)** radiation **(c)** conduction **(d)** refraction

12. The handles of cooking utensils are made of plastic or wood because they are: 1

(a) shiny **(b)** good conductors
(c) heavy **(d)** poor conductors of heat

13. A sundial cannot be used: 1

(a) in the morning **(b)** in the afternoon
(c) in summer **(d)** at night or on cloudy days

14. A bus covers 200 km in 4 hours. Its average speed is: 1

(a) 50 km/h **(b)** 25 km/h **(c)** 800 km/h **(d)** 60 km/h

15. Which organ secretes bile? 1

(a) Pancreas **(b)** Stomach **(c)** Liver **(d)** Gall bladder

16. The system which carries nutrients and oxygen is the: 1

(a) respiratory system **(b)** digestive system
(c) circulatory system **(d)** nervous system

17. Which gas do plants take in during respiration? 1

(a) helium **(b)** oxygen **(c)** hydrogen **(d)** nitrogen

18. The food made by photosynthesis is first in the form of: 1

(a) protein **(b)** vitamin **(c)** glucose **(d)** fat

19. Mirrors in a periscope are placed at: 1

(a) 30° to the tube **(b)** 90° to the tube **(c)** 0° to the tube **(d)** 45° to the tube

20. An extra day is added every four years, making a: 1

(a) light year **(b)** lunar year **(c)** leap year **(d)** solar year

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 15–25 words each.

21. State two properties of acids and two of bases. 2

22. What is an electric circuit? 2

23. State two methods of preventing rusting. 2

24. State two myths about menstruation and correct them. 2

25. What is radiation? How does heat reach us from the Sun? 2

26. What is the SI unit of time and speed? 2

27. What is digestion? Name the organs of the alimentary canal in order. 2

28. What is the role of chlorophyll? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Why can pure water not be used to test for conduction in the same way as lemon juice? 3

30. What does ignition temperature tell us about fuels? Compare paper and kerosene. 3

31. Why do we use clinical thermometers? State precautions. 3

32. How does the structure of villi help absorption? 3

33. Explain with an example how the shape of a shadow can differ from the object. 3

34. Why are the equinoxes called so? 3

35. Why is excess use of fertilisers harmful to soil? 3

OR

A field has become acidic. Name the substance a farmer should add and give the reason. 3

36. Describe the experiment on the role of water in plant growth. 3

OR

What would happen if all stomata were blocked? 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–120 words each.

37. (i) How do we prevent corrosion? 5
(ii) Explain three methods.
(iii) Why is prevention important?

OR

(i) Explain how oxides of metals and non-metals differ. 5
(ii) How do you test them?
(iii) Give one example of each.

38. (i) Why is a ‘balanced social life’ necessary? 5
(ii) How to use technology responsibly?
(iii) Give two examples.

OR

(i) List five points for making adolescence a joyful experience. 5
(ii) Explain two.

39. **Case study:** Two cars start together. Car A moves at 72 km/h; Car B at 15 m/s. 5
(i) Whose speed is greater?
(ii) Distance covered in 1 h by each.
(iii) Who is ahead after 1 h?
(iv) By how much?

OR

Case study: A distance–time table: 0 s 0 m; 10 s 50 m; 20 s 100 m; 30 s 160 m. 5
(i) Is it uniform?
(ii) Speed for first 20 s.
(iii) Speed for last 10 s.
(iv) Overall average speed.

40. **Case study:** Torch light directed at a smooth mirror and a rough wall. 5
(i) Which gives clear image?
(ii) Why?
(iii) Name the reflections.
(iv) Give an everyday example.

OR

(i) A boy stands 2 m from a plane mirror. How far is his image from him? 5
(ii) He moves 1 m closer. New distance?
(iii) State why.

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END OF QUESTION PAPER **Total: 80 Marks**

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